ASCIA

SMART HOME SYSTEMS

SPECIFICATIONS

ASCIA OS

Complete Home Automation System — Platform, OS & Ecosystem

Version	1.0 — Initial Document
Date	February 2026
Status	Confidential — Internal Use
Author	ASCIA Product Team
Based on	Home Assistant OS (Fork + ASCIA Layer)
Target deployment	Dedicated Server / Mini-PC ASCIA (x86_64 / ARM)
Target customers	Premium Residential, Hospitality, Commercial

Table of Contents

SECTION 01

General Presentation

Vision, Objectives & Strategic Positioning

1. General Presentation & Strategic Vision

1.1 Who is ASCIA?

ASCIA is a Quebec-based company specializing in high-end home automation. It distinguishes itself through a fully vertically integrated approach: proprietary hardware design, deployment of a custom home automation operating system, and turnkey installation for its residential, hotel, and commercial clients.

ASCIA covers the entire home automation spectrum: smart lighting, security and alarm, access management (locks, gates, garages), motorized blinds and curtains, multiroom sound system and home cinema, video surveillance, water leak detection with automatic valves, robot vacuum cleaners, lawnmowers and swimming pools, and all everyday IoT devices (coffee machine, snow blower, animal feed dispenser, pressure sensors, etc.).

1.2 The Product: ASCIA OS

ASCIA OS is ASCIA's proprietary home automation operating system. Based on Home Assistant OS as a robust technical foundation, it constitutes a completely rebranded, enhanced and optimized platform, distributed as a pre-installed disk image on an ASCIA server provided to the customer.

ASCIA OS's positioning is radically different from Home Assistant: where HA targets enthusiasts and developers, ASCIA OS targets the non-technical end customer who wants a solution that works from the first connection, without network configuration, without a line of code, without an IT technician.

1.3 Strategic Objectives

- To offer the best home automation experience on the Quebec and Canadian market, accessible to all types of customers without technical expertise
- Create a 100% proprietary ecosystem: ASCIA hardware + ASCIA software + ASCIA support
- Generate recurring revenue through maintenance subscriptions, updates, and proactive remote support
- Drastically reduce deployment and configuration time for ASCIA technicians
- Differentiating ASCIA from all competitors with a unique 3D interface, advanced local AI, and an unparalleled Plug & Play system

1.4 Scope of This Document

This specification covers the entire ASCIA OS system: the operating system, the backend, the web frontend, all functional modules, supported protocols, technical architecture, and the deployment process. The development of the Flutter ASCIA mobile application is covered in a separate document.

1.5 Delivery Model

Component	Description	Provided by
ASCIA Server	Mini PC or pre-configured rack server with pre-installed ASCIA OS	ASCIA
PoE switch	12/24/48 port switch depending on the size of the installation	ASCIA
ASCIA Devices	mmWave sensors, PoE relays, blind/curtain actuators, cameras, audio modules	ASCIA
ASCIA OS	Pre-installed disk image, remote update	ASCIA
Mobile Application	Flutter app for iOS and Android for remote control	ASCIA
Facility	ASCIA Technician — cabling, initial configuration, customer training	ASCIA
Proactive support	Daily/weekly remote diagnostics with client agreement	ASCIA

1.6 Ideal Business Process

Step 1 — Prospecting & Submitting

An ASCIA technician visits the client's premises with the ASCIA POS system. They tour the property, document each room and area, list the desired appliances, and generate a detailed quote. This quote already includes the names of the rooms, areas, and appliances—data that will be used for the automatic configuration of ASCIA OS.

Step 2 — Logistics & Preparation

The validated submission is sent to the logistics manager who orders the parts and prepares the ASCIA server. The server is pre-configured with the submission data (zone names, expected device types) before being shipped to the customer.

Step 3 — Plug & Play Installation

The customer (or technician) connects: modem → ASCIA switch → ASCIA server + all devices to the switch. That's it. ASCIA OS starts, automatically creates the secure network ecosystem, discovers and integrates all devices, and the customer accesses their interface from their phone or browser.

Step 4 — Guided Setup

Upon initial access, a setup wizard guides the client through creating or importing the 3D plan, assigning zones, and validating device names. Basic automations are already created and ready to activate. The client can customize everything via the visual interface.

Step 5 — Continuous Support

ASCIA proactively monitors each installation via its remote diagnostics system. Problems are detected and often resolved before the client even notices them. In the event of an unresolved issue, a ticket is automatically created and the ASCIA team contacts the client.

SECTION 02

Technical Architecture

Architecture Decisions, Stack & Justifications

2. Technological Context & Architectural Decisions

2.1 Why Home Assistant as a Base?

Home Assistant is the world's most mature open-source home automation platform. It represents 8 years of development, 3,000+ contributors, and over 3,000 device integrations. Building ASCIA OS from scratch would require a team of 30+ developers working for 5 years to reach a similar level.

Home Assistant's Apache 2.0 license explicitly allows the creation of commercial products based on its code. ASCIA uses HA as its engine and integration layer, while building its own value-added layer on top: interface, UX, advanced automations, network security, and local AI.

2.2 Selected Approach: Hybrid Layer 1+2+3

After analyzing the alternatives (complete fork, add-ons only, development from scratch), the chosen approach is a hybrid with 3 superimposed layers:

Layer	Technology	Role	Editable by ASCIA
Layer 1 — BONE	HAOS (Home OS Assistant)	Basic Linux system, Supervisor, containers	No (updated via HA)
Layer 2 — Add-ons ASCIA	Docker containers ASCIA	ASCIA Differentiating Features	Yes — development ASCIA
Layer 3 — Frontend ASCIA	SvelteKit (replaces Lovelace)	Full ASCIA branded interface	Yes — development ASCIA

Layer 1 — Unmodified HAOS

HAOS manages booting, hardware, basic networking, and container monitoring. All HA security updates are applied automatically. The HA Core backend runs as is, with all its integrations.

Layer 2 — ASCIA add-ons (supervised Docker containers)

Each differentiating ASCIA feature is packaged as an add-on in a private ASCIA repository. The network-manager add-on handles automatic network isolation. The device-manager add-on handles auto-discovery and onboarding. The ai add-on handles the chatbot and local AI. And so on.

Layer 3 — Frontend SvelteKit ASCIA

The full ASCIA interface is a SvelteKit Single Page Application that completely replaces Lovelace (HA's default frontend). It communicates with HA Core via WebSocket and REST APIs, and with ASCIA add-ons via their own APIs. The client never sees Lovelace—only the ASCIA interface.

2.3 Stack Technique

Component	Technology	Justification
basic operating system	Home Assistant OS (HAOS)	Maturity, ecosystem, Apache 2.0 license
Backend	Home Assistant Core (Python)	3000+ integrations, robust WebSocket API
Container runtime	Docker (via HA Supervisor)	Isolation, standardized deployment
Web frontend	SvelteKit + TypeScript	Performance, SSR, minimum bundle size
3D Rendering	Three.js / Babylon.js	Native WebGL, GPU-accelerated, large community
Local database	SQLite (HA) + TimescaleDB (add-on)	Real-time history + temporal data
MQTT Broker	Mosquito (HA add-on)	IoT standard, high performance
Zigbee	Zigbee2MQTT (add-on)	Maximum compatibility, 3000+ devices
NVR / Cameras	Frigate (add-on)	GPU object detection, universal ONVIF/RTSP
Local AI LLM	Ollama + Whisper (add-on)	100% local, optional GPU, privacy
Object recognition	Frigate + YOLO	Real-time, GPU-accelerated
VPN	WireGuard (add-on)	Performance, simplicity, modern security
Multiroom audio	Music Assistant (add-on)	Compatible with Spotify, Deezer, Sonos, etc.
Network management	NetworkManager + nftables	VLAN, firewall, automatic isolation
CI/CD	GitHub Actions + Private Docker Hub	Automated build, OTA deployment
Monitoring	Prometheus + Grafana (internal)	Server metrics for remote diagnostics

2.4 Overall Architecture Diagram

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ASCIA OS (Disk Image)

LAYER 3 — FRONTEND ASCIA (SvelteKit)
| 3D Dashboard | Cameras | Automations | Network Configuration

| WebSocket + REST
▼
LAYER 2 — ASCIA ADD-ONS (Docker containers) network-mgr |
| device-mgr | nvr/frigate | ai | uvb | audio
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│                    │ Internal HA API                                │
│   ┌────────────────▼──────────────────────────────────┐          │
│                 LAYER 1 — HAOS + HA CORE (Python) State Machine │   │   │
│   │      Automations | 3000+ Integrations                     │   │
│   └───────────────────────────────────────────────────┘          │
└──────────────────────┬──────────────────────────────────┘
                       │ ASCIA PoE Switch
   ┌─────────┬──────────┼───────────┬──────────┐ │
             │          │           │          │
Devices     Cameras      Sensors         Modules      Modem
  PoE/Wi-Fi   ONVIF/RTSP   Zigbee/mmWave   Relay        Internet
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SECTION 03

MODULE 01

Automatic Secure Infrastructure & Network

MODULE 01 — Infrastructure & Automatic Secure Network

Add-on: ascia-network-manager | Priority: P0 Critical

3.1 Problem Solved

Today, properly securing an IoT network requires network administration skills (VLANs, firewalls, segmentation). No ASCIA customer should need to call a network professional or modify their modem. ASCIA OS creates and manages everything automatically.

3.2 Plug & Play Operation

On the first boot of ASCIA OS, the ascia-network-manager add-on analyzes the network environment and automatically configures all network security without any intervention:

1. Detects the client's existing modem and network
2. Creation of a dedicated IoT VLAN (VLAN 100 by default) isolated from the main home network
3. Creation of a separate Wi-Fi IoT SSID (ASCIA-IoT) with an automatically generated unique password
4. Firewall rule configuration (nftables): IoT devices cannot initiate connections to the home network
5. Implementation of an internal ASCIA DNS for device name resolution
6. Configuring internal DHCP with reserved ranges per device type
7. Setting up network intrusion detection (basic IDS rules)

3.3 Dynamic & Adaptive Management

- Automatic detection of each new device connected to the network (MAC address, fingerprinting)
- Immediate application of safety rules on each new device without intervention
- If a device is removed, its rules are automatically deleted.
- Continuous network monitoring: alerts in case of abnormal device behavior
- Detection of unauthorized access attempts between the IoT network and the home network
- Weekly network report: list of devices, bandwidth, incidents

3.4 Hardware Compatibility

Scenario	Configuration	Support
Client with modem basic (Bell, Videotron)	ASCIA OS manages everything downstream of the modem	Complete
Customer with Unifi (Ubiquiti)	UniFi Controller API integration for advanced VLAN management	Complete
Client with Omada (TP-Link)	Omada Controller API Integration	Complete
Client with router advanced (pfSense, OPNsense)	Configuration via API or guided manual mode	Partial
Client with switch manageable	802.1Q VLAN automatically configured via SNMP	Complete

3.5 Network Management Interface

- Network dashboard: all connected devices with IP address, MAC address, VLAN, status, ping, bandwidth
- Network map: graphical visualization of connections between devices
- Firewall rule management via visual interface (no command line required)
- VLAN management: create, modify, delete VLANs from the interface
- Network history: log of all connections, disconnections, anomalies
- Smart network alerts with natural language explanations: 'Your Wi-Fi is weak in the living room — consider adding an access point'
- Internet outage report: 'Your internet connection is offline. Local devices continue to function. Netflix and Spotify are unavailable.'

SECTION 04

MODULE 02

Dynamic Real-Time 3D/2D Interface

MODULE 02 — Real-Time Dynamic 3D/2D Interface

Add-on: ascia-3d-engine | Frontend: SvelteKit + Three.js | Priority: P0 Critical

4.1 Vision

The 3D interface is the core differentiator of ASCIA OS. Rather than a list of devices or a grid of buttons, the customer sees their actual home in 3D and interacts directly with the devices in their physical location. It's the most intuitive experience possible: no need to search for where a device is, simply look at the 3D representation of your home.

4.2 Creation of the Plan

Import from third-party software

- SweetHome 3D (.sh3d): direct import with automatic conversion to ASCIA 3D format
- SketchUp (.skp): Import and conversion to 3D ASCIA format
- GLB/GLTF format: direct import of existing 3D models
- ASCIA rendering targets the visual quality of SweetHome 3D with superior interactivity and fluidity.

Creation in the ASCIA interface

- 2D drag & drop editor: draw the parts, size them, name them
- Automatic generation of the 3D model from the drawn 2D plan
- Library of standard house models as a starting point

Scan LiDAR from the mobile app

- The customer scans their home with their compatible iPhone Pro or Android device
- The 3D model is automatically uploaded to ASCIA OS
- (Full details in the ASCIA mobile application specifications)

4.3 Real-Time Dynamics

Every change in the state of a device is instantly reflected in the 3D model. The states are received via WebSocket from HA Core and applied to the 3D rendering with a target latency of less than 100ms.

Device / Sensor	Behavior in 3D	Technical Details
Door (sensor) opening)	The door opens/closes visually	Animation of rotation on hinge, actual angle if available
Window (sensor)	The window opens/closes	Animation translation or rotation depending on type
Blind / Curtain (actuator)	The blind goes up or down	Animation proportional to the percentage of opening
Light ON/OFF	The light comes on with a bright halo	WebGL light emission, intensity depending on the dimmer
RGB Light	Exact color reflected with color projection	HSB color calculation → WebGL light color
Colored light variable	Color temperature reflected (warm/cold)	Kelvin temperature → WebGL color
FP2 Presence Sensor (Aqara)	Silhouette or colored dot in the room	X/Y position in the room according to sensor data
Multi-presence	N silhouettes according to the number of people detected	Real-time tracking, silhouette movement
Water leak	Area of concern highlighted in pulsating blue + alert	Animated WebGL overlay on the detected area
Vibration sensor	A pulsating icon on the device in question	CSS animation on a 3D overlay
Coffee machine	IActive cone + animated steam when on	Lightweight WebGL Particle System
Robot vacuum cleaner	Represented in its actual position with trajectory	Robot card extraction → overlay on 3D plan
Heat map	Overlay color gradient based on temperature sensors	Spatial interpolation between sensors (simplified kriging)
Alarm triggered	Pulsed red zone, alert icons	High-visibility WebGL overlay

4.4 Interaction & Control from 3D

- Clicking on a device in the 3D view opens a contextual control panel
- Direct control from the 3D model: turn on/off, adjust brightness, change RGB color
- Drag and drop devices onto the map to reposition them (edit mode)
- Long press on an area to access area settings
- Zoom and rotate using touch gestures or mouse
- Lock mode: lock 3D navigation to prevent accidental movements
- Smooth and animated switching between 3D view and 2D view (floor plan)

4.5 Dynamic Emergency Dashboard

In the event of a critical incident, the 3D interface automatically transforms to display only the information and controls relevant to the emergency. This behavior is triggered by ASCIA automations and alert sensors.

Incident	Dashboard Behavior	Displayed Controls
Water leak detected	Automatic zoom on the main valve in 3D	Close valve / Call plumber / All water sensors
Fire detection / smoke	Area affected is shown in red, evacuation plan overlay	Alarm / Firefighters / Fire Doors
Intrusion / Alarm	Full-screen cameras of the area	Alarm / Police / Camera Access
Power outage	Devices powered off are grayed out, active UPS is highlighted	Generator set status / Backup
Server failure	Alert banner + restart guide	Restart ASCIA / Contact support

4.6 Zones & Automatic Assignment

- Creating a zone: the ASCIA technician draws it in 3D by defining the perimeter
- For each connected device, ASCIA requests its zone of affiliation.
- The device is automatically placed in the 3D environment in the correct area with the corresponding icon.
- All dashboards, groups, and automations related to this area are updated automatically.
- The zone is propagated throughout the system: Alexa, Google Home, reports, history

4.7 Target Performance

KPI	Target	Condition
3D frame rate (desktop)	60 FPS constant	Average house plan (5-8 rooms, 50 units)
Framerate 3D (mobile via WebView)	30-60 FPS	iPhone 12 Pro / Pixel 6
Latency update status → 3D	< 100ms	Local WebSocket, simple entity
3D plan loading time	< 2 seconds	Floor plan converted from SweetHome 3D, browser cache
GPU Memory	< 256MB	Optimized 3D model, compressed textures

SECTION 05

MODULE 03

Device Self-Discovery & Onboarding

MODULE 03 — Self-Discovery & Device Onboarding

Add-on: ascia-device-manager | Priority: P0 Critical

5.1 Principle

Any device connected to the ASCIA network (via cable or wirelessly) is automatically detected, identified, and integrated into ASCIA OS without any manual intervention, code entry, or technical configuration. The only thing the client needs to do is name the device and assign it a zone.

5.2 Discovery Protocols

Protocol	Discovery Method	Time AVERAGE	AppAffected areas
PoE (cable) network)	LLDP+ Detection MAC ASCIA fingerprint	< 5 seconds	ASCIA wired cameras, relays, and sensors
Wi-Fi	mDNS + SSDP + UPnP + DHCP fingerprint	< 15 seconds	Standard Wi-Fi devices
Zigbee	Zigbee2MQTT permit join	Instant After activation	All Zigbee devices
Z-Wave	Z-Wave JS inclusion mode	Instant After activation	All Z-Wave devices
KNX	ETS scan of the KNX bus	< 30 seconds	Wired KNX devices
ONVIF / RTSP (cameras)	Scan network ports ONVIF (80, 8080, 2020)	< 20 seconds	All standard IP cameras
Bluetooth BLE	Continuous BLE scan	< 10 seconds	BLE sensors, beacons
Matter / Thread	Matter commissioning	< 30 seconds	Modern Matter devices
MQTT devices third party)	MQTT Auto-discovery (topic homeassistant/#)	Instant	Any MQTT-compatible device
ASCIA Devices owners	MAC ASCIA profile + device certificate	< 3 seconds	All ASCIA equipment

5.3 Onboarding Process

8. The device connects to the ASCIA network
9. ascia-device-manager detects it via the appropriate protocol
10. The system identifies the type of device (camera, door sensor, light, relay, etc.)
11. For ASCIA devices: full profile applied automatically (mode, default configuration)
12. For third-party devices: the system suggests the detected type; the customer confirms or corrects it.
13. The system asks for: zone of affiliation (selection from the list of created zones)
14. The system automatically generates a name based on the zone + the type: 'Room 1 Door Sensor'
15. The device appears in 3D at the position of its zone, with the corresponding icon.
16. The corresponding group is updated: 'Bedroom 1 Lights' automatically includes the new light
17. Relevant automation blueprints are proposed for activation

5.4 Simplified Button Pairing

Zigbee Pairing Mode

- The technician clicks on 'Activate Zigbee Pairing' in the ASCIA interface
- The Zigbee coordinator enters inclusion mode (configurable duration: 30s, 60s, 120s)
- All Zigbee devices within range that are in pairing mode appear in a list
- For each device: type selection → area selection → automatic naming → 3D placement

PoE Pairing Mode

- Any PoE device connected to the ASCIA switch is automatically detected
- For ASCIA devices: instant setup without any action required
- For third-party devices: the interface displays 'New PoE device detected — Port 12 — Configure'

NFC Pairing mode (mobile application)

- Scan the QR code or NFC tag of an ASCIA device from the mobile app
- Instant addition and configuration without any menu navigation

5.5 Intelligent Automatic Nomenclature

The system automatically generates consistent and intuitive device names based on ASCIA submission and zone data. This naming convention is applied uniformly throughout the interface, automations, voice commands, and reports.

- Convention: [Type] [Area] [Index if multiple] — Ex: 'Living Room Light 1', 'Front Door Sensor', 'Garage Camera'
- The names are derived from the areas created in the ASCIA submission.
- Any name change is automatically propagated throughout the system.
- Export of the complete list of names and entities available at any time
- The names are compatible with voice recognition (Alexa, Google Home, ASCIA AI)

SECTION 06

MODULE 04

Protocol Management

MODULE 04 — Visual Protocol Management

Visual interfaces for KNX, Zigbee2MQTT, PoE/Wi-Fi, Omada/Unifi | Priority: P0

6.1 KNX Interface

KNX is a high-end wired protocol used in premium commercial and residential installations. Its configuration is notoriously complex. ASCIA radically simplifies this experience.

- Visual mapping of KNX group addresses: graphical representation of the bus and devices
- Import of the ETS file (KNX project) for automatic configuration
- KNX command test interface: send a command and see the response in real time
- KNX bus diagnostics: status of each device, communication errors, signal quality
- KNX zone configuration aligned with ASCIA zones (no double configuration)
- KNX bus monitoring: load graph, collision detection

6.2 Zigbee2MQTT Interface

- Visualization of the Zigbee mesh map: graphical topology with link quality (LQI)
- Status of each Zigbee device: online, offline, battery, firmware version
- Over-the-air (OTA) firmware update for all compatible Zigbee devices in bulk
- Advanced configuration per device: bindings, reporting intervals, calibration
- Adding additional Zigbee coordinators to extend coverage
- Automatic detection and resolution of mesh problems (orphan nodes, poor LQI)
- Zigbee router: automatic suggestion to add repeaters if coverage is insufficient

6.3 PoE & Wi-Fi Interface

- View by switch: list of all ports with connected devices, PoE power consumption, VLAN
- View by device: IP address, MAC address, ping latency, bandwidth, connection status, online time
- Enabling/disabling a PoE port from the interface (remote reset)
- Alert if a device consumes more PoE power than expected (possible fault)
- ASCIA IoT Wi-Fi Management: List of connected devices, signal strength, disconnections
- Bandwidth per device: detect energy-intensive devices on the network

6.4 Omada & Unifi Interface

- Connecting to an existing Omada or Unifi controller via API

- Visualizing the complete network topology (APs, switches, routers) in ASCIA
- VLAN management from the ASCIA interface without going through the native controller
- Monitoring Wi-Fi coverage with signal heat map
- ASCIA alerts based on Omada/Unifi events (AP failure, overload)
- Automatic configuration of ASCIA IoT VLANs on UniFi/Omada switches

6.5 Global Mapping Interface

A unified view groups all devices, protocols, entities, and their network status into a single interface. This is ASCIA's network administration dashboard.

- Hierarchical tree: Zones → Device types → Devices → Entities
- Quick filters: show only offline devices, low battery, errors
- Bulk actions: rename, move zone, update firmware on multiple devices
- Export of the complete inventory (CSV/PDF) for client documentation

SECTION 07

MODULE 05

Native NVR, Cameras & Surveillance

MODULE 05 — Native NVR, Cameras & Smart Surveillance

Add-on: ascia-nvr (based on Frigate) | Priority: P0 Critical

7.1 Automatic Camera Integration

Any camera connected to the ASCIA network is automatically discovered and integrated, without any manual input of streams or URLs. Maximum compatibility guaranteed.

Protocol	Compatibility	Automatic Discovery
ONVIF	All cameras compatible with ONVIF (98% of the market)	Automatic network scan
RTSP	Direct RTSP stream on any compatible device	Fingerprint detection
HTTP/MJPEG	Basic cameras with MJPEG stream	Scan standard ports
PoE ASCIA	ASCIA Proprietary Cameras	Automatic Full Profile
Reolink	Optimized native integration	Reolink API
Hikvision	Optimized native integration	Hikvision API
Dahua	Optimized native integration	Dahua API
Axis	Optimized native integration	Axis API

7.2 Native NVR System

Registration

- Continuous 24/7 recording to local storage (internal SSD or network NAS)
- Event-driven recording: triggered by motion detection, presence, or sensor
- Scheduled recording: define recording time slots per camera
- Configurable quality and resolution per camera
- Camera-configurable retention period: 1 week to 12 months

Navigation through history

- Timeline scrubbable for navigating through records
- Calendar view: see which days have recordings
- Event-based navigation: jump directly to the moments when something was detected
- Snapshot at any time from the interface with local save + gallery
- Exporting video clips as MP4 from the interface

7.3 Intelligent Search in Videos

The most differentiating feature of the ASCIA NVR: contextual search which cross-references sensor data with video recordings.

- 'Show me all the entries and exits from yesterday' → ASCIA takes the door sensor history and automatically returns clips from 30 seconds before and after each trigger
- 'Who rang the doorbell this week?' → Doorbell camera clips + sensor log
- 'Show me the last time the garage was opened' → clip of the exact moment
- Search by keywords, by area, by device, by event type, by time range
- Search by detected object: 'Show me all the clips where a car is detected at the entrance'

7.4 Frigate & Advanced Detection

Object detection

- Real-time detection: people, cars, bicycles, animals, parcels
- GPU acceleration: if a GPU is present in the ASCIA server, Frigate uses it automatically
- Detection zones configured directly from the ASCIA 3D plan (no native Frigate configuration)
- Area masks: exclude parts of the image (moving tree, busy road)
- Configuring sensitivity by object and by area

Facial Recognition

- Optional activation of facial recognition (explicit customer consent required)
- Local face database — no data sent to external servers
- Automatic association: 'Jean-Pierre returned at 6:42 PM'
- Personalized notification based on the recognized person
- Facial profile management interface from the ASCIA interface (no direct Frigate configuration)

Camera Notifications

- Push notification with image thumbnail at the time of detection
- Contextual notification: 'An unidentified person is at the entrance' with preview
- Doorbell notification: thumbnail + 'Talk' button for the intercom
- False alarm reduction: AI filter to eliminate trivial alerts

SECTION 08

MODULE 06

Local Artificial Intelligence

MODULE 06 — Local Artificial Intelligence

Add-on: ascia-ai | Stack: Whisper + Ollama + local LLM | Priority: P1

8.1 Philosophy

ASCIA's AI is 100% local. No data from the home (voice commands, questions, device statuses) ever leaves the ASCIA server. It's an AI built for a specific home, learning the habits of its occupants, and becoming more relevant over time.

8.2 Conversational Features

Request	AI Response	Data Source
'Is the oven on?'	'Yes, the kitchen oven has been on for 23 minutes at 180°C'	State entity four
'How many people are at home?'	'3 people detected: living room, bedroom 1, office'	mmWave presence sensors
'Turn off all the lights in the living room'	Execution + 'All 4 living room lights are off'	HA call service
'What is the security status?'	Full report: armed alarm, active cameras, sensors	Multiple States Entities
'Close all the blinds'	Execution on all actuators + confirmation	Group call service
'Create an automation...'	Guided dialogue to create automation	Automation engine
'What's the temperature outside?'	'It's -3°C outside with snow forecast for this afternoon'	Weather sensor or API
'Show me the entrance cameras'	Automatic opening of the camera feed in the interface	NVR ASCIA

8.3 Creating Automations with AI

One of the most useful features for reducing ASCIA technician deployment time. The client or technician describes the desired automation in natural language, and the AI creates it.

- 'I want all the living room lights to turn on when I get home' → automation created
- 'Turn off the heating when a window is open' → automation created with confirmation
- For complex automations, AI asks up to 10 clarifying questions

- Each AI-created automation is presented in natural language for validation before activation.
- AI suggests variations or improvements to the proposed automation
- Creation by voice or text in the ASCIA chatbot

8.4 Intelligent Reports & Suggestions

- Automatic or on-demand morning report: 'Last night: 3 motion detections outside, Jean-Pierre returned home at 6:42 PM, the oven was automatically switched off at 9:15 PM'
- Report dictated via the house speakers or displayed on the dashboard
- Proactive suggestions: 'It's raining and your office window has been open for 20 minutes. Would you like me to close it?'
- Automation suggestions based on detected patterns: 'You always turn off the kitchen lights after 11pm. Would you like me to do it automatically?'
- Daily security report available upon request

8.5 Infrastructure & Performance

Component	Technology	Configuration
Voice transcription	OpenAI Whisper (local)	Mod small.en or medium model depending on available RAM
Conversational LLM	Ollama with Llama 3.1 / Mistral model	GPU accelerated if present, otherwise CPU (latency ~2-4s)
Acknowledgement facial	Frigate + InsightFace (local)	GPU highly recommended
TTS (Text-to-Speech)	Piper TTS (local)	Natural speech synthesis, personalized ASCIA voice
Memory conversational	local SQLite	History of the last 100 exchanges per user
Learning habits	HAS log analysis	Detection of recurring patterns

Target latency: less than 2 seconds for common commands without GPU, less than 500ms with GPU.

8.6 AI Interface

- Chatbot accessible from all pages of the ASCIA interface (floating button)
- Microphone in the web interface and from all microphones connected to the home
- Text response within the interface AND voice response via the speakers in the relevant area
- Activation via configurable (disableable) wake word 'Ascia'
- Viewable conversation history
- Silent mode: text-only replies

SECTION 09

MODULE 07

Presence & Location

MODULE 07 — Presence & Localization (mmWave, UWB, BLE)

Proprietary ASCIA sensors + third-party integrations | Priority: P1

9.1 Proprietary mmWave ASCIA Sensors

ASCIA produces its own mmWave presence sensors, inspired by the Everything Smart Home Presence Sensor One format, but with proprietary improvements. Each sensor integrates two transmitters: one for static detection (person sitting, asleep) and one for dynamic position tracking.

Sensor features

- Static presence detection: person who is motionless, asleep, meditating
- Dynamic tracking: X/Y position of the person in the room
- Counting the number of people in the area
- Heart rate detection (optional, depending on the model)

Configuration via the ASCIA 3D interface

- Delineating detection zones directly in the 3D plane (drag & draw)
- Zone exclusion: eliminate false alarms (fan, pet)
- Guided calibration: the interface indicates how to position the sensor for optimal coverage
- Real-time visualization of the detected position in the 3D plane
- Sensitivity thresholds can be adjusted via the interface (no YAML configuration required).

9.2 UWB (Ultra-Wideband) System

The UWB ASCIA system allows centimeter-level localization of each inhabitant in the house, triggering automations based on the specific person and their exact position.

- Installation of UWB ASCIA anchors in the house (minimum 3-4 anchors for complete coverage)
- UWB badge or key fob provided to each resident
- Real-time visible position in the 3D plane: each person represented by their avatar
- Person-based automations: 'When Jean-Pierre enters the kitchen → turn on the coffee maker (his preferred coffee profile)'
- Accuracy: < 30cm indoors
- Configuring UWB zones from the ASCIA 3D interface

9.3 BLE (Bluetooth Low Energy) System

- A less expensive alternative to UWB, with part accuracy (vs. centimeter accuracy for UWB)
- Detection via the customer's phone or dedicated BLE beacons
- Compatible with Apple AirTag, Tile, and generic BLE tags
- ASCIA OS offers two options: BLE (economical) and UWB (premium)

9.4 Presence-Based Automation

Trigger	Automation Example
Jean-Pierre enters the kitchen	Coffee maker → Jean-Pierre coffee profile, kitchen light 80%
Sophie enters the room after 10 p.m.	Lights dimmed 20%, blinds down, reduced temperature
Empty house (all members absent)	Away mode activated, alarm armed, energy saving
First to arrive at home	Home mode activated, entry lights, normal heating
Child returns from school	Parent notification, snack automation (coffee maker, kitchen light)
Unrecognized person in secure area	Security alert, notification, camera recording

SECTION 10

MODULE 08

Automation, Blueprints & Autopilot

MODULE 08 — Automations, Blueprints & Autopilot Mode

Enhanced HA Core layer + ascia-automation-engine add-on | Priority: P0

10.1 Automation Engine

ASCIA OS retains and enhances the Home Assistant automation engine—the most powerful on the open-source home automation market. ASCIA's goal is to make this engine accessible to non-technical users while preserving its power for complex installations.

10.2 Automatic Creation of Groups & Blueprints

One of the major sources of time savings for ASCIA technicians is the automatic creation of groups and blueprints as soon as a device is added.

Action	Automatic ASCIA Creation
Add Light to Living Room 3	Added to the 'Living Room Lights' group, blueprint presence → suggested lights
Add Store Room 1	Added to the 'Blinds Bedroom 1' group, blueprint schedule + weather available
Add Motion Sensor Hallway	Blueprint 'Movement → Corridor Light' created, ready to activate
Add Kitchen Water Sensor	Blueprint 'Leak → Close Valve + Notification' created
Add Living Room Thermostat	Blueprint energy saving + presence integration proposed
Add Entry Lock	Blueprint options: 'Arrival → Unlock', 'Departure → Lock'

Automatically created automations are inactive by default. The client or technician activates and customizes them.

10.3 Automation Log & Timeline

- Each automation trigger is logged: time, trigger, condition checked, action performed, result
- Visual timeline: a chronology of all the day's events on a timeline
- Filter by automation, by device, by zone, by event type
- Option to replay an automation from the log for testing
- Alert if an automation fails repeatedly

10.4 Scenes

- Simplified scene creation: select the devices + their desired state, name them
- ASCIA predefined scenes: Morning, Departure, Return, Evening, Cinema, Night, Away, Vacation
- Activating scenes from the 3D interface, dashboards, voice AI, Alexa/Google Home
- Scenes created automatically according to the patterns defined during submission

10.5 Autopilot Mode

Autopilot mode lets ASCIA OS make all home automation decisions autonomously, based on configured rules and learning habits.

- Basic rules configured by the ASCIA technician during installation
- ASCIA decides: which lights to turn on depending on presence and time, when to arm the alarm, how to manage the heating
- Continuous learning: the system refines its decisions based on customer feedback
- Always modifiable: the client can override any decision at any time
- Transparent mode: every autopilot decision is logged and explained in the log

10.6 Advanced Presence Simulation

- In Away mode, ASCIA replicates the client's habits to simulate a believable presence
- Based on local machine learning: analysis of logs from the last 30 days
- Random variations to avoid predictable patterns
- Configurable: choose which devices participate in the simulation
- Automatic activation in Vacation mode

10.7 Calendar Mode

- Connecting to family members' Google Calendars
- Event-triggered automations: 'Meeting at 8am → wake-up at 6:45am if client is still asleep'
- Integrated management of Quebec and Canadian public holidays
- Birthdays: voice reminder + festive lighting
- The system learns calendar-related routines and anticipates them automatically.

10.8 Route Management

- The customer registers their usual addresses (work, school, grocery store)
- Waze/Google Maps API integration: travel time estimation with real-time traffic
- 'Your meeting starts at 9am, the journey takes 42 minutes. I'll wake you up at 7:45am'
- Smart departure notification: 'It's time to leave to arrive on time'
- Automation return: 'Jean-Pierre arrives in 12 minutes → preheat, entrance lights'

SECTION 11

MODULE 09

Home Multimedia, Audio & Cinema

MODULE 09 — Multimedia, Audio & Home Cinema

Add-on: ascia-audio (based on Music Assistant) | Priority: P1

11.1 Multiroom Audio System

- Complete music management in all areas of the home
- Zone grouping: playing the same music in multiple rooms simultaneously
- Supported sources: Spotify, Deezer, Apple Music, Web Radio, local files, YouTube Music
- Speaker compatibility: Sonos, Denon HEOS, Bang & Olufsen, ASCIA Audio devices, Chromecast Audio, AirPlay
- Full controls: play/pause/skip/shuffle/repeat/volume/zone EQ
- Overall house volume: single slider controlling all speakers
- Audio programming: 'Wake me up at 7am with Jazz in the bedroom at 30% volume'
- AI voice announcements on speakers: reports, alerts, reminders

11.2 Home Cinema & Multimedia

- Simplified integration: Apple TV, Chromecast, Fire TV, Android TV, projectors
- Management via the ASCIA interface: Netflix, Prime Video, Disney+, YouTube, TV channels
- Cinema Scene in one click: dim lights, lower blinds, projector on, sound configured
- Control multiple devices simultaneously from a single interface
- Integration of Quebec content providers: Club illico, Vrai, Tou.tv
- Universal remote control from the ASCIA interface
- Multimedia-related automations: 'When the movie starts → lower lights to 15%'

SECTION 12

MODULE 10

Security, Access & User Management

MODULE 10 — Security, Access & User Management

High Availability Layer + Ascia Security Add-on | Priority: P0 Critical

12.1 User & Profile Management

Profile	Access	Restrictions
Administrator	Full access to the entire system	None
Family Member	Access to all domestic areas	No system configuration
Children's Fashion	Areas defined by the parents	No access cameras, locks, alarm
Guest	Areas and devices specifically authorized	Limited time, no settings
Provider	Service areas (e.g., garage) only	Limited time offer, full action log
ASCIA Technician	Temporary full technical access	Session recorded, full logs
ASCIA Support	Remote diagnostic access	Only with explicit customer agreement, temporary

12.2 Remote Access

Option 1 — ASCIA Cloud

- Easy access from anywhere without configuration
- End-to-end TLS 1.3 encryption
- Optimized latency via ASCIA relay servers in Canada

Option 2 — WireGuard VPN (total local)

- Automatic VPN configuration on first boot
- Certificates generated and managed automatically by ASCIA OS
- The client activates the VPN from the interface — no network configuration required.
- VPN client management: create, revoke, time limit
- No data passes through third-party servers

Remote ASCIA support

- Temporary access granted by the client: 15 min / 30 min / 1 hour / 4 hours
- The session was recorded in its entirety in the audit log.

- Revocable at any time by the customer

12.3 Connections & Audit Log

- Each system connection recorded: user, date/time, IP address, geographic location, device
- Each failed login attempt was logged with the same information
- Alerts: 'Attempted connection from an unusual location (Paris, France) — Was that you?'
- Export of the audit log to CSV for enterprise clients
- Newspaper retention: minimum 1 year

12.4 Encryption & Key Management

- TLS 1.3 for all remote communications
- Self-renewing SSL certificates (Let's Encrypt or ASCIA PKI)
- AES-256 encryption for sensitive data stored locally
- Professional key management: automatic rotation, centralized revocation
- Weekly automated security audit with report
- Detection of expired or compromised certificates

12.5 Privacy Mode

- Each zone, room or device can activate a 'Privacy' mode
- Private area → blurred area in the 3D plan
- Global privacy mode → disables all cameras and microphones in one click
- Fine-tuning: choose which devices are disabled in private mode
- History of privacy mode activations
- Guest privacy mode: private areas are automatically hidden for guest users

SECTION 13

MODULE 11

Energy, Monitoring & History

MODULE 11 — Energy, Monitoring & History

Add-on: ascia-analytics | Priority: P1

13.1 Energy Monitoring

- Consumption dashboard by device, by zone, by day/week/month/year
- Ranking of the most energy-intensive appliances
- Estimated monthly cost based on the Hydro-Québec rate (or personalized rate)
- Comparison with previous periods and seasonal averages
- Clear graphics accessible to a non-technical client
- Overconsumption alerts: 'Your consumption this month is 30% higher than normal'

13.2 Energy Saving Automation

- Pre-designed blueprints: automatic light shut-off when no one is present, adaptive heating
- Integration of differentiated tariffs (peak/off-peak hours Hydro-Québec)
- Automatic economy mode: ASCIA adjusts heating and air conditioning according to the outside weather
- AI Suggestions: 'Your air conditioner has been running for 6 hours while it's 16°C outside — should you turn it off?'

13.3 Complete History & Event Log

ASCIA OS records absolutely everything that happens in the house. Nothing is lost, everything can be recovered.

- Each change of state of each device is recorded with a precise timestamp.
- Each automation trigger: trigger, conditions checked, actions executed, result

- Each connection, disconnection, access attempt
- Each alert, notification, report generated
- Configurable retention: 1 month to 1 year
- Full-text search in the newspaper: 'Show me everything that happened in the kitchen last night'
- Visual timeline: chronology of the day with all events
- Export the log to CSV/PDF for archiving

SECTION 14

MODULE 12

Notifications, Reports & Events

MODULE 12 — Notifications, Reports & Events

Integrated into ascia-core + ascia-ai | Priority: P0

14.1 Notification System

Kind	Canals	Example
Critical alert	Mobile push notification + loud sound + 3D alert	Incfire, water leak, intrusion
Notification with action	Push mobile with action buttons	'Oven on for 2 hours — [Turn off] [Leave on]'
Notification with image	Mobile push notification + camera thumbnail	'Motion detected at the entrance' + preview
Notification doorbell	Push button + vignette + intercom	'Something 'One at the door' + image + [Speak]
Report system	Email + ASCIA interface	Weekly General Status Report
Battery alert	Push + badge in the interface	'Office door sensor — battery 10%'
Network alert	IASCIA Interface + optional push	'Offline device: Garage Camera'

14.2 Notification Configuration

- Activation/deactivation by device, by event type, by user
- Quiet hours: suspension of non-critical notifications at night
- Configurable trigger thresholds: 'Only alert me if the battery is below 15%'
- Grouping notifications: avoid flooding with alerts for the same event
- Repeated unacknowledged critical alerts
- User-based routing: some notifications are only for the admin, others for everyone

14.3 Reporting System

- Automatic morning report: summary of the night, system status, notable events
- On-demand security report: all detections, connections, alerts
- Monthly energy report: consumption, estimated cost, comparison
- Weekly system report: status of all devices, batteries, storage, latency

- AI-dictable report: 'Ascia, give me the night report' → voice reading on speakers
- Full configuration: choose the devices and events included in each report

14.4 Structured Event System

- Doorbell Event: Full-screen camera thumbnail + intercom + history
- Alarm Event: 3D in red, area cameras, emergency contacts
- Water leak event: 3D zoom on valve, water sensors, plumber's number
- Recognized Visitor Event: 'Jean-Pierre has arrived' with visit history
- Each event is logged with full context to easily trace the history

SECTION 15

MODULE 13

Storage, Backup & Data Management

MODULE 13 — Storage, Backup & Data Management

Add-on: ascia-storage-manager | Priority: P0

15.1 Storage Management

- Storage dashboard: used/available space by category (videos, logs, backups, music)
- Proactive alerts: 80% → warning, 95% → critical alert with suggested action
- Automatic archiving: move old recordings to NAS or external storage
- Supports network NAS (Synology, QNAP, Unraid), internal SSD, external USB
- Data retention period by type: configurable by the client from 1 month to 1 year
- Automatic compression of old data to optimize space

15.2 Backup System

Backup Type	Frequency	Content	Destination
Backup configuration ASCIA	Daily	Configuration, automations, users, zones	Local + Encrypted Cloud
Full system backup	Weekly	Configuration + historical database	Local + Encrypted Cloud
Backup video	Continuous (NVR)	Camera recordings	Local (SSD/NAS)
Monthly backup archiving	Monthly	Complete system snapshot	Encrypted cloud
Backup before update	With each update	System state before the update	Local

- One-click restore from any backup point
- Success or failure notification for each backup
- Automatic backup integrity test (verification that restoration would work)

15.3 Update Management

- Manual mode: the client decides when to apply each update
- Programmable automatic mode: weekly, monthly, or annual
- Detailed release notes displayed before installation
- Automatic backup before each update
- Automatic rollback if the update causes a detected regression
- Update channel: stable (recommended), beta (ASCIA testers)

SECTION 16

MODULE 14

Diagnostics, Self-Healing & Remote Support

MODULE 14 — ASCIA Diagnostics, Self-Healing & Remote Support

Add-on: ascia-diagnostics | Priority: P0 for ASCIA customers

16.1 Automatic Proactive Detection

Condition Detected	Alert	Automatic Action
Device battery < 15%	Customer notification	Replacement suggestion
Storage > 80%	Interface alert	Suggestion for archiving or extension
Device offline > 5 min	Customer notification	Ping attempt + service restart
ASCIA server overheating	Critical alert	Reduced load + ASCIA alert support
Abnormal house temperature (>35°C)	Customer alert	Suggestion for ventilation, heating check
High network latency (>500ms)	Interface alert	Automatic network diagnostics
SSL certificate expires in 30 days	Admin alert	Automatic renewal attempted
Backup failed	Admin alert	New attempt + notification if repeated failure
Outdated device firmware	Information	Updated OTA proposal

16.2 Self-Healing — Automatic Recovery

Before alerting the customer or creating a support ticket, ASCIA OS attempts to resolve the problem itself, in a transparent and documented manner.

18. Problem detected → automatic diagnosis to identify the cause
19. Applying the first remedy: restarting the service in question
20. If the problem persists → remedy 2: reset the module configuration
21. If the problem persists → remedy 3: complete restart of the add-on
22. If still present → customer notification with clear explanation
23. If critical and unresolved → automatic creation of an ASCIA support ticket
24. Sending the ticket with a complete diagnosis to the ASCIA team
25. ASCIA contacts the customer, intervenes remotely, or sends a technician.

16.3 On-Demand Self-Diagnosis

- A complete test that can be launched at any time or scheduled
- Testing each device: ping, latency, status, command response
- Automation testing: checking triggers and actions
- Network test: bandwidth, latency per device, Wi-Fi quality
- Server test: CPU, RAM, temperature, disk space
- Comprehensive diagnostic report with health score (0-100) and recommendations

16.4 ASCIA Tele-Diagnosis

With the client's explicit consent, ASCIA OS sends a diagnostic report to the ASCIA team daily or weekly. This allows for proactive intervention before the client notices a problem.

- The customer activates/deactivates remote diagnostics from the interface with a single click
- Data sent: device status, battery levels, errors, server metrics (no personal data, no images)
- The ASCIA team can identify and correct problems remotely
- Automatic ASCIA alert if a critical indicator is detected at a client site
- Revocable at any time, without impacting the system's operation

SECTION 17

MODULE 15

Interface, Personalization & Customer Experience

MODULE 15 — Interface, Personalization & Customer Experience

Frontend SvelteKit ASCIA | Priority: P0

17.1 Design Principles

- 100% ASCIA branded interface — no mention of Home Assistant visible
- Mobile-first: designed primarily for smartphones, adapted for tablets and desktops
- Perceived latency < 100ms for all local interactions
- Accessibility: WCAG AA contrast, adaptive text sizes
- Multilingual support: French (priority), English

17.2 Advanced Customization

- Full drag and drop: reposition any dashboard element
- Right-click → context menu with quick options on each item
- Visual themes: light, dark, customized (colors, client logo)
- Multiple layouts: grid, list, 3D, cameras, energy
- Dedicated editing mode: modify the interface without risk of triggering commands
- Favorites and custom shortcuts at the top of the interface
- Keyboard shortcuts for power users

17.3 Responsive & Multi-Platform

Platform	Experience	Notes
Smartphone (via app Flutter)	Optimized native mobile interface	ASCIA Flutter Application
Mobile browser	Responsive web interface	Access without an app required
Tablet (iPad, Android)	Extended interface with sidebar + content	2-column layout
Desktop (browser)	Full interface with all panels	Resizable windows
Dedicated wall screen	Fullscreen kiosk mode	Permanent 3D Dashboard
Apple Watch	watchOS app (via Flutter app)	Scenes and quick controls
Android Wear OS	Wear OS app (via Flutter app)	Scenes and quick controls

SECTION 18

MODULE 16

External Integrations

MODULE 16 — External Integrations

Google Home, Alexa, Calendars, External APIs | Priority: P1

18.1 Google Home & Amazon Alexa

- Expose ASCIA devices to Google Home and Alexa in just a few clicks from the interface
- Creating custom phrases for Alexa voice commands
- Alexa routines created by the customer are automatically translated into ASCIA automations
- Two-way synchronization: a change in ASCIA is reflected in Alexa/Google Home
- ASCIA interface for managing exposure: choose which devices are visible to each assistant
- Support for Google Assistant Actions for advanced contextual commands

18.2 System Integrations

Integration	Functionality	Priority
Google Calendar	Event-based automations, reminders, wake-up	P1
Waze / Google Maps	Estimated route, return time, departure alerts	P1
Weather (Weather.com / Weather Canada)	Weather automation, suggestions, reports	P0
Hydro-Québec	Actual rates, peak hours, energy optimization	P1
ASCIA POS System	Import submission for automatic configuration of names/zones	P0
ASCIA Support Platform	Automated ticketing, remote diagnostics, remote access	P0
Apple HomeKit	Exposing ASCIA devices to HomeKit (via HA Bridge)	P2
SmartThings	Integration of existing SmartThings devices	P2

SECTION 19

Business Process

ASCIA Client Deployment & Workflow

19. Sales Process & Customer Deployment

19.1 Complete Workflow

Stage	Actor	Action	ASCIA Tool
1. Lead	Commercial ASCIA	Qualifying the prospect, scheduling an appointment	CRM ASCAI
2. Submission	Technician ASCIA	On-site visit, documentation of areas and equipment	ASCIA POS System
3. Validation	Director reader commercial	Submission validation and signature	POS ASCIA
4. Logistics	Director logistics	Ordering parts, preparing the server	ASCIA ERP
5. Preconfiguration	Technician / Developer ASCIA	Import submission → ASCIA OS preconfigures names and zones	HSASCIA OS Admin
6. Delivery	Logistics	Delivery of the server + switch + devices to the customer's premises	—
7. Installation	Client or technician	Physical connection: modem → switch → server + devices	—
8. First startup	ASCIA OS	Network auto-configuration, device discovery, onboarding	ASCIA OS
9. Configuration guided	Client (assisted)	Create/import 3D plan, assign zones, names	ASCIA OS + App
10. Training	Technician ASCIA	Presentation of the interface, scenes, and AI	—
11. Support continuous	ASCIA Team	Tproactive remote diagnostics, updates, incidents	ASCAI Support

19.2 Target Deployment Times

Scenario	Installation Size	Objective Time Deployment
Apartment / Condo	10-20 devices	Full day (8 hours) ASCIA technician
Single-family home	20-50 devices	1-2 days ASCIA technician
Luxury home	50-100 devices	2-4 days, 2 technicians

Scenario	Installation Size	Objective Time Deployment
Boutique Hotel	100-500 devices per floor	1-2 weeks, dedicated team
Commercial building	500+ devices	Custom project

SECTION 20

Technical Architecture

Server Specifications, APIs & Data

20. Detailed Technical Architecture

20.1 ASCIA Server Specifications

Component	Minimum Configuration	Configuration Recommended	Configuration Premium
Processor	Intel Core i5 (8th gen+) / AMD Ryzen 5	Intel Core i7 (12th gen) / AMD Ryzen 7	Intel Core i9 / AMD Ryzen 9
RAM	16 GB DDR4	32 GB DDR4	64 GB DDR4
Storage system	256 GB NVMe SSD	512 GB NVMe SSD	1 TB NVMe SSD
Data storage	1 TB HDD / SSD	4TB SSD	8+ TB RAID
GPU (AI/Frigate)	None (CPU only)	NVIDIA RTX 3060 / AMD RX 6600	NVIDIA RTX 4080+
Network	1 Gbps Ethernet	2.5 Gbps Ethernet	10 Gbps Ethernet
Ports	4x USB-A, 1x HDMI	6x USB-A, 2x HDMI, 2x DP	According to chassis
Food	65W Adapter	500W ATX Power Supply	750W Gold ATX Power Supply
Format	Mini-PC (Intel NUC-like)	mini-ITX tower	ATX Tower / 1U Rack

20.2 ASCIA Add-ons — Complete List

Add-on	Technical Base	Main Function	Resources
ascia-network-manager	NetworkManager + nftables	Network isolation, VLAN, automatic firewall	lightweight CPU
ascia-device-manager	Python custom + HA API	Self-discovery and device onboarding	lightweight CPU
ascia-3d-engine	Node.js + API WebSocket	Sreal-time 3D engine server	average CPU
ascia-nvr	Frigate + Go2RTC	NVR, object detection, video streams	GPU heavily recommended
ascia-ai	Ollama + Whisper + Piper	Local LLM, voice transcription, TTS	GCOULD recommended
ascia-audio	Music Assistant	Multiroom audio management	lightweight CPU
ascia-presence	Custom + Zigbee2MQTT	mmWave, UWB, BLE management	lightweight CPU

Add-on	Technical Base	Main Function	Resources
ascia-analytics	TimescaleDB + Grafana	Advanced history, energy, reports	CPU/RAM AVERAGE
ascia-vpn	WireGuard	Secure remote access	lightweight CPU
ascia-diagnostics	Python + Prometheus	Monitoring, self-healing, remote diagnostics	lightweight CPU
ascia-backup	Rclone + custom	Local and cloud backup management	lightweight CPU
ascia-automation-ai	Python + LLM	Create cars using natural language	Depends ascia-ai

20.3 APIs Exposed by ASCIA OS

API	Protocol	Authentication	Use
HA WebSocket API	WebSocket	Token Bearer	ASCIA Frontend, mobile app
HA REST API	HTTPS	Token Bearer	One-off operations
ASCIA Device API	REST	ASCIA Token	Onboarding proprietary devices
ASCIA 3D API	WebSocket	ASCIA Token	Real-time updates to the 3D model
ASCIA AI API	REST + WebSocket	ASCIA Token	Chatbot, voice commands
ASCIA NVR API	REST + HLS	ASCIA Token	Video streams, history, snapshots
ASCIA Analytics API	REST	ASCIA Token	History, reports, energy
MQTT Broker	MQTT/MQTTS	Username/Password	ASCIA and third-party IoT devices

SECTION 21

Roadmap

Development Phases & Milestones

21. Roadmap & Development Phases

Phase 0 — Preparation (Month 0, 4 weeks)

- Setting up the development environment (macOS + Linux for HAOS)
- Fork of HAOS and creation of the private ASCIA OS repository
- CI/CD setup: GitHub Actions → build image → test server deployment
- Creation of the ASCIA design system: colors, typography, basic components
- Definition of the internal ASCIA API (interface contract between add-ons and frontend)
- Operational development ASCIA server with basic HAOS

Phase 1 — MVP Foundations (Months 1-4)

DELIVERABLE: ASCIA OS installable and testable by the ASCIA technical team

Module	Description	Weeks
Frontend SvelteKit base	Replacing Lovelace with the ASCIA frontend (WS login, basic pages)	1-4
ascia-network-manager v1	Automatic network isolation, VLAN, basic firewall	2-6
ascia-device-manager v1	Self-discovery: PoE, Zigbee, Wi-Fi, ONVIF	3-8
Automatic nomenclature	Naming system based on zones and device types	5-8
Basic NVR (ascia-nvr v1)	Integrated Frigate, continuous recording, stream viewing	4-10
3D Interface v1	Basic 3D rendering with real-time states (lights, doors, blinds)	6-14
Backup System v1	Daily backup configuration + restore	8-12
Testing & Stabilization	Integration testing, critical bug fixes	12-16

Phase 2 — Advanced Features (Months 5-9)

DELIVERABLE: First pilot installation at an ASCIA customer site

Module	Description	Weeks
Local AI (ascia-ai v1)	Chatbot, Whisper voice commands, local LLM (Ollama)	17-24

Module	Description	Weeks
3D Interface v2	Full dynamics (RGB, presence, robots, thermal)	17-22
mmWave v1 Presence	Integration of mmWave ASCIA sensors, 3D mapping	19-26
Auto-blueprints & groups	Automatic creation of groups and blueprints when adding devices	20-24
Protocol interfaces	KNX, Zigbee2MQTT, PoE/Wi-Fi, Omada/Unifi visuals	22-28
Self-healing v1	Proactive detection, automated remediation, support tickets	24-30
Advanced multi-user	Child, guest, and provider profiles; granular permissions	26-30
Report & Events	Full log, timeline, configurable reports	27-32
Pilot Tests	Deployment at 3-5 ASCIA pilot clients	32-36

Phase 3 — Commercial Platform (Months 10-15)

DELIVERABLE: ASCIA OS v1.0 — Available to all ASCIA customers

- Autopilot mode and presence simulation
- AI v2: automation suggestions, morning report, wake word
- Complete UWB system with 3D visualization and per-person automation
- Comprehensive remote diagnostics and ASCIA support portal
- Google Home, Alexa, Google Calendar, and Waze integrations
- Complete network management interface (Omada, UniFi)
- Advanced energy monitoring with Hydro-Québec integration
- Local facial recognition (opt-in)
- Flutter mobile app v1.0 (App Store + Google Play deployment)

Phase 4 — Continuous Innovation (Months 16-24)

DELIVERABLE: ASCIA OS v2.0 — Reference Platform

- Public ASCIA SDK for third-party integrators
- Multi-property mode: manage multiple installations from one interface
- AI v3: Deep learning of habits, advanced predictions
- Full Matter/Thread integration for next-generation devices
- Hotel module: multi-room management, automatic check-in/check-out
- Public ASCIA API for certified third-party developers

Phase	Duration	Effective Dev	Main Deliverable
Phase 0 — Preparation	4 weeks	2 devs (senior full-stack + DevOps)	Infrastructure & basic tools
Phase 1 — MVP	4 months	3 devs (2 full-stack + 1 DevOps)	ASCIA OS testable
Phase 2 — Advance	5 months	4-5 dev (3 full-stack + 1 AI + 1 DevOps)	Customer pilot

Phase	Duration	Effective Dev	Main Deliverable
Phase 3 — Commercial	6 months	5-6 developers + 1 QA + 1 PM	ASCIA OS v1.0
Phase 4 — Innovation	9 months	6-8 developers + 1 QA + 1 PM	ASCIA OS v2.0

SECTION 22

Team & Resources

Profiles, Tools & Budget Estimates

22. Team, Resources

22.1 Required Profiles

Role	Key Skills	Entry phase	Critical
Full-Stack Lead Developer (Senior)	Python/HA, SvelteKit/TypeScript, Docker, Linux	Phase 0	Critical
Senior Full-Stack Developer	SvelteKit, WebSocket, Three.js/Babylon.js (3D), API design	Phase 0	Critical
Dev Backend / Add-ons (Mid-Senior)	Python, Docker, MQTT, IoT protocols (Zigbee, KNX, PoE)	Phase 1	Critical
AI/ML Developer (Mid-Senior)	Python, Ollama/LLM, Whisper, Frigate, GPU computing	Phase 2	High
DevOps / Infrastructure (Mid)	GitHub Actions, Docker, Linux, HAOS, network	Phase 0	High
Flutter Mobile Developer (Senior)	Flutter, Dart, native Swift, ARKit, UWB	Phase 2	High
UI/UX Designer (Freelance)	Figma, systems design, home automation interface	Phase 0 (partial)	Average
QA Engineer (Mid)	Automated testing, IoT testing, real-world devices	Phase 2	Average
Product Manager	Roadmap, specs, team coordination	Phase 1	Average

SECTION 23

KPIs & Quality

Non-Functional Requirements & Performance Indicators

23. Non-Functional Requirements & KPIs

23.1 Performance

KPI	Target	Measure	Condition
Local command latency	< 100ms	From click to execution on the device	Local network, simple order
Command latency distance	< 500ms	Via WireGuard VPN	10 Mbps+ connection
3D interface framerate (desktop)	60 FPS constant	standard desktop GPU	8-room plan, 100 entities
Loading the 3D plan	< 2 seconds	First display after login	Warm cover
ASCIA OS startup time	< 90 seconds	From server startup to accessible interface	Server configuration minimal
Discover PoE devices	< 5 seconds	From the connection to the appearance in ASCIA	Switch ASCIA
ONVIF camera discovery	< 20 seconds	De network connection to ASCIA detection	Rlocal network
Camera video stream latency	< 500ms	HLS local, resolution 1080p	Local network 100 Mbps
AI response (command) (voice)	< 2s without GPU, < 500ms with GPU	From the end of the sentence to the action	Model small.en Whisper
ASCIA OS Crash Rate	< 0.01% / month	No restarts planned	By installation
Local system availability	99.5%	Not to mention breakdowns Internet	Per month

23.2 Security

Requirement	Standard	Implementation
Encryption communication	TLS 1.3 minimum	Let's Encrypt + PKI ASCIA
Authentication	Tokens + optional 2FA	HA Auth + ASCIA Auth Layer
IoT network isolation	802.1Q VLAN	ascia-network-manager

Requirement	Standard	Implementation
Audit log	Minimum 1-year retention	Encrypted SQLite + cloud archiving
Key management	Automatic rotation 90 days	ascia-key-manager
Data compliance	GDPR + Quebec's Bill 25	Privacy by design
Pen test	Annual + before each major version	External service provider

23.3 Reliability & Maintenance

- ASCIA OS must operate 24/7 without intervention for a minimum of 6 months
- Update without service interruption for non-critical add-ons
- ASCIA Core update with reboot < 60 seconds
- Restoration from backup in < 10 minutes for configuration
- Complete technical documentation for each ASCIA add-on
- Stable ASCIA API with semantic versioning (no breaking changes without a migration guide)

SECTION 24

Appendices

Glossary, Licenses & Technical References

24. Annexes

24.1 Glossary

Term	Definition
HAOS	Home Assistant Operating System — The basic Linux system on which ASCIA OS is built
HA Core	Home Assistant Core — Python backend managing entities, states, and automations
ASCIA add-on	HAOS-supervised Docker container, adding proprietary ASCIA features
Lovelace	Home Assistant's default frontend interface, replaced by the SvelteKit ASCIA frontend
Entity	Logical representation of a device or attribute in HA (e.g., living room light = entity light.livingroom)
Blueprint	Configurable and reusable automation template in Home Assistant
VLAN	Virtual LAN — A virtual local area network that allows IoT devices to be isolated from the home network.
MQTT	Message Queuing Telemetry Transport — Standard lightweight IoT messaging protocol
ONVIF	Interoperability standard for IP cameras — used for camera self-discovery
RTSP	Real-Time Streaming Protocol — A video streaming protocol used by IP cameras
Frigate	Open-source NVR with integrated GPU object detection — used as the ASCIA NVR engine
Zigbee2MQTT	Gateway that converts Zigbee devices into MQTT/HA compatible entities
WireGuard	Modern VPN protocol — fast, secure, simple — used for remote ASCIA access
GLB/GLTF	Web standard binary 3D file format used for ASCIA 3D models
Whisper	OpenAI's open-source speech transcription model — used locally in ASCIA AI
Ollama	Local LLM server allowing language models to run on the ASCIA server
mmWave	Millimeter-wave radar technology used for static and dynamic presence detection

Term	Definition
UWB	Ultra-Wideband — Indoor Precise Location Radio Technology (< 30cm)
PoE	Power over Ethernet — Power supply via network cable, standard for wired ASCIA devices
KNX	High-end industrial wired protocol for building automation
HLS	HTTP Live Streaming — An adaptive video streaming protocol used for webcam feeds
Self-Healing	The system's ability to automatically detect and correct its own faults
Plug & Play	Ability to operate immediately after plugging in without manual configuration
SvelteKit	Compiled JavaScript framework used for the ASCIA frontend (Lovelace replacement)

24.2 Licenses & Legal Compliance

Component	License	Constraint for ASCIA
Home Assistant OS	Apache 2.0	Commercial use permitted, as indicated in the credits.
Home Assistant Core	Apache 2.0	Commercial use permitted, as indicated in the credits.
Frigate	MIT	Commercial use permitted without major restrictions
Zigbee2MQTT	GPL v3	Use as a separate service/add-on — compliant
Music Assistant	Apache 2.0	Commercial use permitted
WireGuard	GPL v2 (kernel) Linux)	Integrated into the Linux kernel, compliant usage
Mosquitto MQTT	EPL v2 + EDL	Commercial use permitted
Ollama	MIT	Commercial use permitted without restriction
Whisper (OpenAI)	MIT	Commercial use permitted — local model
Three.js	MIT	Commercial use permitted without restriction
SvelteKit	MIT	Commercial use permitted without restriction

Note: This table is for informational purposes only. A full legal review is recommended before the commercial launch of ASCIA OS.

ASCIA Smart Home Systems

Specifications — ASCIA OS — Version 1.0